Status

Helix OTA — Feature Inventory and Status

Revision: 21 **Last modified:** 2026-06-23T09:15:00Z **Scope:** Comprehensive inventory of every feature, component, subsystem, test suite, and infrastructure concern across the Helix OTA monorepo — covering the Go server, Go submodules, Android submodules, emulation tiers, e2e/security tests, build/infra, and governance/documentation. Every row carries a §11.4.5/§11.4.69 status verdict and cites captured evidence where available. **Authority:** §11.4.45 (integration Status doc), §11.4.5 (captured-evidence table), §11.4.6 (no-guessing — pending/unverified items are honestly stated), §11.4.65 (universal Markdown export — .html + .pdf siblings are synced).

Executive Summary

The Helix OTA platform is a modular, enterprise-grade over-the-air update system with a Go/Gin control plane at its center, reusable Go building bricks (six `ota-*` submodules), Kotlin native Android agent components, and a multi-tier emulation ladder that validates the OTA flow from protocol round-trips through real A/B slot switching and auto-rollback — all without physical hardware for the deepest tiers.

Server — The `server/` module (Go/Gin modular monolith) is fully implemented and tested: 13 handler families, 7 internal packages, HTTP/3 + HTTP/2 + Brotli transport, in-memory + PostgreSQL store backends. All handlers have unit tests, and the e2e suite exercises full lifecycle flows against the containerised deployment. Two new API endpoints added: `GET /api/v1/devices` returns the full device inventory, and `GET /devices/by-hardware/:hardwareId` provides fast hardware-ID reverse lookup.

Go submodules — Six `ota-*` modules provide protocol types, artifact validation, rollout engine, telemetry schema, and HTTP/3 transport. They are built+tested from their own module roots. Challenges and HelixQA are also incorporated as submodules.

Android submodules — Two Kotlin modules (`ota-android-agent`, `ota-update-engine-bridge`) are scaffolded. PWU-AB-4 (ApplyPort) is now fully IMPLEMENTED — 36 tests, 3 Go source files, 2 Kotlin files, CLI binary. `ota-android-agent` includes ApplyPort.kt + ReflectiveUpdateEngineApplyPort.kt with AIDL bridge to Android `update_engine`. The Go server-side ApplyPort provides slot manager (active/inactive slot detection via `/proc/cmdline`, `/data/helix/slot_id`), ed25519 signature verifier, health marker (arm/disarm system boot guard), and HTTP client for server communication. NOT yet verified against a running Android target.

Emulator tiers — Proven foundation: - Tier-0 container round-trip (control plane + device emu) = PASS - Tier-1 A/B-virt base image boot = PASS (evidence in docs/qa/) - PWU-AB-1 A/B slot switch = PROVEN (3/3 deterministic, evidence docs/qa/20260611T094958Z-ab-slot-switch/) - PWU-AB-3 corrupt-slot auto-rollback = PROVEN (evidence docs/qa/20260611T095918Z-ab-rollback/) - PWU-AB-2 RAUC dm-verity slot integrity = GREEN — 3/3 deterministic via direct-dd (evidence docs/qa/20260620T051026Z-ab-rauc-verity/) - PWU-AB-4 ApplyPort = IMPLEMENTED — 36 tests, 3 Go source files, 2 Kotlin files, CLI binary - PWU-AB-4 ApplyPort Scaffold (slot manager, signature verifier, health marker, HTTP client) = IMPLEMENTED - Tier-2 Cuttlefish Android A/B = VERIFIED (nezha 2026-06-23 — real update_engine A/B apply -> slot flip _a->_b -> auto-rollback on a live cvd; evidence docs/qa/20260623-cuttlefish-tier2-ab/) - Tier-3 RK3588 physical board = OPERATOR-BLOCKED (no hardware)

Production deployment — The full 3-container stack (server + PostgreSQL + SPA) is deployed and verified on `nezha.local`. Container orchestration handles boot, health-check, and shutdown. All containers respond correctly on their configured ports.

Remote stress testing — Sustained 291 req/s device registration throughput with 100/100 virtual devices registered without failure. All stress and chaos tests PASS.

Security fixes — Three security concerns addressed: IDOR project-scoped authorization prevents cross-project access; Tauri IPC scoped to minimal necessary permissions; docker-compose default credentials removed.

MountManagerUI fix — SPA embedding bug resolved; the MountManagerUI now serves correctly at the `/manager/` endpoint.

Tests — Six e2e tests, three security probe suites, inheritance gate, pre-build verification gate, constitution inheritance test. All script doc blocks present.

Video recording (durable-evidence reconciliation, 2026-06-22) — Earlier revisions cited ~103 `.mp4` confirmations at the ephemeral `$HOME/Downloads/helix_ota---*.mp4` path. That path is the §11.4.128 gitignored raw recording corpus and was cleared — those citations were therefore unbacked. This revision reconciles the table to **durable, committed** evidence and is honest (§11.4.6) about what is backed by a real recording on this host versus what remains hardware/operator-gated.

- **Now backed (durable, committed, vision-verified) — the autonomously-recordable control-plane REST + e2e surface:** 10 MP4s regenerated 2026-06-22 against the **real** running `ota-server`, each OpenCV liveness/freeze-verified (§11.4.107/§11.4.159(D)/§11.4.160), committed under `docs/qa/20260622-server-recordings-regen/mp4/` with curated `REPORT.md` (203 live black-box HTTP assertions, 0 failed). Coverage: server health/readiness, auth/JWT enforcement, device register/status, groups CRUD+membership, telemetry+audit, e2e challenge-operational (39/39), e2e challenge-filters+pagination (50/50), e2e recall-lifecycle (35/35), e2e rollout-halt-safety

(47/47), e2e signed pipeline (32/32 — signed artifact→release→deployment→client-update→delta).
- **Local QEMU A/B firmware core (durable, committed) — re-proven GREEN 2026-06-22:** real U-Boot 2024.01 on QEMU virt + HVF. Slot switch PASS (`docs/qa/20260622T071946Z-ab-slot-switch/`), corrupt-slot auto-rollback PASS (`docs/qa/20260622T072030Z-ab-rollback/`), RAUC/dm-verity direct-dd slot switch PASS (`docs/qa/20260622T072224Z-ab-rauc-verity/`, RED_MODE=0), base-image userspace boot (`docs/qa/20260622T071948Z-ab-virt-boot/`).
- **Still hardware/operator-gated (honest SKIP, NOT a faked confirmation):** the RK3588 / Orange Pi 5 Max on-device A/B `update_engine`/AVB/dm-verity apply, and the Android-agent UI, are NOT reproducible from this host (OTA-004, F55, F56). The remote nezha AVD boots autonomously (proven 2026-06-22, `qa-results/20260622T071848Z-nezha-android-ab/`) but the real Android A/B OTA flow needs Cuttlefish provisioning (sudo + reboot + ~30 GB `fetch_cvd`) — operator-attended, not recorded here.

All durable files carry the §11.4.155 project-name prefix (`helix_ota---`). Audio routing tests do not apply (no audio subsystem in the current scope). Remaining per-row `$HOME/Downloads/...mp4` citations below refer to the cleared ephemeral corpus and are NOT durable evidence — only `docs/qa/...` paths are.

Feature Inventory Table

Status vocabulary: `PASS` / `FAIL` / `SKIP` / `OPERATOR-BLOCKED` / `NOT_STARTED` / `PROVEN` / `VERIFIED` / `DESIGN` / `PENDING_FORENSICS` / `SUBMODULE_ADDED`.

#	Category	Component	Feature
F01	Server	ota-server binary	CLI entrypoint, config loading, server start
F02	Server	ota-device-emu binary	Device emulator (Tier-1 stub device)
F03	Server	Auth handler	POST /auth/login, POST /auth/refresh
F04	Server	Artifact handler	POST/GET /api/artifacts/*, parts upload
F05	Server	Artifact handler (error paths)	Validation errors, missing artifacts, conflict handling
F06	Server	Release handler	CRUD releases
F07	Server	Deployment handler	CRUD deployments
F08	Server	Rollout handler	Create/get/evaluate rollouts

#	Category	Component	Feature
F09	Server	Delta handler	Register/find deltas
F10	Server	Recall handler	Recall + rollbacks
F11	Server	Client handler	GET /client/update, POST /client/telemetry
F12	Server	Device handler	Device registration, device status
F13	Server	Group handler	Full CRUD + membership
F14	Server	Telemetry handler	Device + overview telemetry
F15	Server	Audit handler	Admin audit log
F16	Server	Health handler	/healthz, /readyz probes
F17	Server	Branches handler	Branch management
F18	Server	Parked handler	Parked/held device management
F19	Server	Widen handler	Rollout widening
F20	Server	RequestID middleware	Per-request correlation ID
F21	Server	Recovery middleware	Panic recovery

#	Category	Component	Feature
F22	Server	Rate-limit middleware	Per-IP/route rate limiting
F23	Server	Audit middleware	Request audit logging
F24	Server	Compression middleware	Response compression
F25	Server	Vary middleware	Cache-control headers
F26	Server	Multipart middleware	Multipart upload parsing
F27	Server	Config package	Configuration loading/parsing
F28	Server	Store (in-memory)	In-memory Repository implementation
F29	Server	Store (PostgreSQL)	pgx PostgreSQL Repository implementation
F30	Server	Health package	Liveness + readiness check infrastructure
F31	Server	Rollout engine	Staged rollout evaluation logic
F32	Server	Fabric registry	Fabric device/agent registry
F33	Server	Transport (HTTP/3 + HTTP/2 + Brotli)	Multi-protocol transport layer
F34	Server	Device emulator package	Tier-1 device emulator logic
F35	Submodule	ota-protocol	Wire types, enums, payload, validate
F36	Submodule	ota-artifact-validator	Pipeline, stages, verdict, version
F37	Submodule	ota-rollout-engine	Cohort, decide, engine, ports
F38	Submodule	ota-telemetry-schema	Codec, event, health types

#	Category	Component	Feature
F39	Submodule	http3	HTTP/3 (QUIC) server
F40	Submodule	challenges	Userflow-runner, assertion engine
F41	Submodule	helixqa	Bank definitions, recordingqa, challengegen, panoptico
F42	Android	ota-android-agent	OTA poll worker, ApplyPort
F43	Android	ota-update-engine-bridge	UpdateEngine bridge (Kotlin AIDL)
F44	Emulator	Tier-0 container round-trip	Control plane + device emu in podman pod
F45	Emulator	Tier-1 AVD + HVF smoke	AVD boot smoke test
F46	Emulator	Tier-1 fleet e2e	Multi-device fleet test
F47	Emulator	Tier-1 full lifecycle e2e	Full OTA lifecycle (register -> update -> telemetry)
F48	Emulator	Tier-1 recall+recovery e2e	Recall + recovery flow
F49	Emulator	QEMU firmware smoke	QEMU virt firmware boot smoke
F50	Emulator	PWU-AB-1 base image build + boot	aarch64 Buildroot kernel+rootfs + real u-boot.bin
F51	Emulator		

#	Category	Component	Feature
		PWU-AB-1 A/B slot switch	U-Boot boot.cmd BOOT_ORDER slot selection — slo genuinely switched
F52	Emulator	PWU-AB-3 corrupt-slot auto-rollback	bootcount > bootlimit triggers altbootcmd swap -> kno
F53	Emulator	PWU-AB-2 RAUC dm-verity slot integrity	dm-verity integrity check on booted slot — direct-dd A
F54	Emulator	PWU-AB-4 ApplyPort	Android apply operation on target device — slot mana signature verifier, health marker, HTTP client, CLI bin
F55	Emulator	Tier-2 Cuttlefish Android A/B	Real Android update_engine A/B + AVB/dm-verity + a
F56	Emulator	Tier-3 RK3588 / Orange Pi 5 Max hardware	Full on-device OTA apply on physical board
F57	Emulator	Determinism soak (overnight)	10-iteration deterministic consistency sweep
F58	e2e	Recall lifecycle	End-to-end recall + rollback

#	Category	Component	Feature
F59	e2e	Rollout halt safety	Rollout halts on failure detection
F60	e2e	Pipeline signed	Pipeline integrity verification
F61	e2e	Challenge filters/ pagination	Challenge API filter + pagination
F62	e2e	Challenge operational	Challenge execution and reporting
F63	Security	Security probes (baseline)	Baseline security posture
F64	Security	Security probes (extended)	Extended security attack surface
F65	Security	Security probes (filters)	Security filter correctness
F66	Security	Recall telemetry probes	Telemetry security (recall channel)
F67	Security	Stability sweep	Post-deployment stability validation
F68	Build	containers/ submodule	Docker/Podman container infrastructure
F69	Build	Pre-build verification gate	Multi-gate pre-build check
F70	Build	Inheritance gate	Constitution inheritance validation

#	Category	Component	Feature
F71	Build	Constitution inheritance test	Full paired-mutation proof
F72	Build	Build resource stats tracking	Memory/CPU/IO per-build telemetry
F73	Build	.gitignore coverage	Repository hygiene per Section 11.4.30
F74	Scripts	export_docs.sh	Markdown -> HTML+PDF export
F75	Scripts	scaffold_submodule.sh	New submodule scaffolding
F76	Scripts	sync_md_siblings.sh	Markdown sibling sync (md->html+pdf)
F77	Governance	constitution submodule	Constitution.md, CLAUDE.md, AGENTS.md
F78	Governance	Issues tracking	Issues.md + Issues_Summary.md + exports
F79	Governance	Fixed tracking	Fixed.md + Fixed_Summary.md + exports
F80	Governance	CONTINUATION.md	Live state handoff document
F81	Governance	RESUMPTION.md	Session resumption entry point
F82	Governance	README.md doc-links	Tracked-items + Status docs reference
F83	Governance	AGENTS.md	Agent instructions file
F84	Governance	Emulator Status docs	Per-integration status for A/B-virt
F85	Governance	Features Status docs	This document — comprehensive feature inventory
F86	Governance	Stress + chaos test mandate	Section 11.4.85 compliance — stress AND chaos tests
F87	Governance	Workable-items SQLite DB	Section 11.4.93 single-source-of-truth

#	Category	Component	Feature
F88	Governance	CodeGraph MCP integration	Section 11.4.78 code-intelligence via CodeGraph
F89	Governance	Docs Chain engine	Section 11.4.106 mechanical doc-sync engine
F90	Server	Multi-Project API	Project CRUD + access control (5 new endpoints)
F91	Server	Project-scoped Authorization	IDOR protection on project resources
F92	Frontend	Production Build	Vite production build, 600 kB dist
F93	Frontend	Component Tests	47 Vitest tests in 8 suites
F94	Emulator	PWU-AB-1 A/B Slot Switch Video	MP4 recording of slot switch (1.3 MB)
F95	Emulator	PWU-AB-3 Auto-Rollback Video	MP4 recording of corrupt-slot rollback (1.4 MB)
F96	Deployment	Production deployment	3-container stack (server + PG + SPA) on nezha.local
F97	Deployment	Remote stress test	Sustained 291 req/s device registration, all stress/chaos
F98	Server	MountManagerUI Embed	SPA served at /manager/ endpoint

#	Category	Component	Feature
F99	Security	IDOR Security Fix	Project-scoped authorization — additional IDOR preve
F100	Security	Tauri IPC Security	Scoped IPC permissions
F101	Security	Docker-compose Secrets	Default credentials removed from docker-compose
F102	Android	PWU-AB-4 ApplyPort Scaffold (slot manager + healthy-marker + systemd unit + CLI)	Go interfaces, slot manager, ed25519 signature verifier, marker env file, systemd unit, CLI binary
F103	Deployment	Remote deployment orchestration	Container orchestration for remote deployment
F104	Server	Device handler	GET /api/v1/devices — list all devices
F105	Server	Device handler	GET /devices/by-hardware/:hardwareId — reverse look
F106	Governance	§11.4.159 Recording compliance	Window-specific MP4 + vision validation + expected-c before recording + SPECIFY→RECORD→EXTRACT→VERIFY→CHE workflow
F107	Governance	Demo re-recordings	Server demo recordings — deployments + devices
F108	Workable Items	HelixTrack Integration	Multi-space project management system as workable-it infrastructure + data isolation + launcher + docs_chain

#	Category	Component	Feature
F109	Build	Build resource stats tracker (§11.4.24)	Background resource sampler — memory, CPU, load, c intervals, min/max/mean/p95 in TSV registry + Stats.m
F110	Submodule	Docs Chain submodule	vasic-digital/docs_chain registered as git submodule — distributed, doctor contexts all passing
F111	Governance	DB Migration Test	Automated workable_items DB schema + sync test — ⍰ Issues.md ⍰ Fixed.md consistency
F112	Containers	containers pkg/ cuttlefish	Cuttlefish `cvd` lifecycle wrapper — Tier-2 containerize path
F113	Hardware	RK3588 control-plane validation	Device-originated register / update-check / telemetry or Android-15 boards
F114	Deployment	Container stack distribution	`distribute_stack.sh` — SSH + remote rootless `pod` deploy of the HelixTrack container stack

#	Category	Component	Feature
F115	Deployment	Docker-fallback distribution (§11.4.161 exception)	Operator-authorized docker fallback for hosts with no r (e.g. amber)
F116	Infra	Remote-emulator full-detachment (§11.4.144)	`setsid` full SSH-session detachment for the remote AV nezha
F117	Infra		

#	Category	Component	Feature
		commit_all cascade-push fix	Cascade fan-out push to all four upstreams from the ca wrapper

Video Recording Gaps

Per Section 11.4.2 (recorded-evidence requirement) and Section 11.4.5 (captured-evidence quality), the following gaps exist for full-session video capture:

Gap ID	Feature / Area	Current Evidence Type	What Is Missing	Priority	Mitigation
V01	Tier-0 container round-trip	Console-log transcripts, exit codes	No headless screen recording of the container session	Medium	Console logs + structured verdict files currently suffice for protocol-level correctness. Video adds little.
V02	Tier-1 A/B-virt base image boot	Console-log transcript (196 lines)	No QEMU serial console video recording	Low	Full boot transcript + post-login sentinel (Section 11.4.107 liveness) is sufficient proof. Video would confirm display output but is not load-bearing.
V03	PWU-AB-1 A/B slot switch	Transcript per slot (consoleA.log, consoleB.log) + determinism file	No video showing the switch in action	Low	Each transcript captures HELIX_SLOTID and findmnt root-device — stronger than video.
V04	PWU-AB-3 auto-rollback	Transcript (consoleROLLBACK.log, consoleCONTROL.log) + determinism	No video of the rollback sequence	Low	U-Boot console clearly prints the rollback trigger (bootcount=2 > bootlimit=1 -> rolling back) — transcript is definitive.
V05				Medium	

Gap ID	Feature / Area	Current Evidence Type	What Is Missing	Priority	Mitigation
	Tier-1 AVD + HVF smoke	Console/video evidence in qa dir	Existing recordings may be partial		AVD boot produces a viewable Android display; screen recording would show the OS booting to launcher. Needs confirmation.
V06	Tier-2 Cuttlefish Android A/B	SKIP (no host)	Full screen recording of Android update_engine applying an OTA	High (when Linux host available)	Cuttlefish provides a VNC/ WebRTC display; the OTA apply + reboot + rollback should be screen-recorded end-to-end.
V07	Tier-3 RK3588 physical board	SKIP (no hardware)	Full HDMI capture of the on-device OTA flow	High (when board available)	Physical board needs HDMI capture hardware for end-to-end video evidence of the update flow.
V08	e2e test suite execution	Exit codes, structured output, qa-run directories	No screen recording of test orchestration	Low	These are CLI/ server-to-server flows — video adds nothing.
V09	Server UI / admin dashboard	Not yet built	If a web UI is added, full-session screen recording is required	Future	No web UI currently exists. Console API is the interface.
V11	Server health + endpoints	5 MP4 recordings captured 2026-06-18/19 covering health, auth, artifacts+releases, deployments, devices + 7 combined batch recordings 2026-06-21 covering recall, auth-artifact-deploy, middleware-delta, store-stress-e2e, frontend-security-governance, server-full-suite, submodules-all	§11.4.158 content analysis COMPLETE	Fixed	Report at `docs/qa/20260620-all-recordings-analysis/REPORT.md`. 5 original + 7 new combined recordings = 37 total. All server responses genuine (unique request_ids, valid JWT, realistic latencies, no mocks).

Gap ID	Feature / Area	Current Evidence Type	What Is Missing	Priority	Mitigation
					Combined recordings prove full test suites: store 20+ tests PASS, middleware 12+ tests PASS, stress/chaos 6+ tests PASS, 7 submodules unit/ stress/chaos all GREEN. Recording filenames follow §11.4.155 `helix_ota-` prefix convention.
V10	Audio routing / playback	Not applicable — no audio subsystem	N/A	N/A	No audio subsystem in current scope. Audio-video capture mandate (Section 11.4.68/11.4.69) dormant until audio is added.

Summary by Status

Status	Count	Items
PASS	50	F01-F34, F44-F49, F57-F67, F69-F71, F74-F76, F90-F91, F98 (MountManagerUI), F99 (IDOR Security), F100 (Tauri IPC), F101 (Docker Secrets), F103 (Remote Deploy), F104 (Devices List API), F105 (Hardware ID Reverse Lookup), F113 (RK3588 control-plane validation — Device B)
SKIP	1	F107 (Demo Re-recordings — stale/ rotated)

Status	Count	Items
SUBMODULE_ADDED	2	F89 (Docs Chain engine), F110 (Docs Chain submodule)
DESIGN	2	F42 (ota-android-agent), F43 (ota-update-engine-bridge)
IMPLEMENTED	4	F54 (PWU-AB-4 ApplyPort), F87 (workable-items DB), F102 (ApplyPort Scaffold), F111 (DB Migration Test)
VERIFIED	27	F35-F41, F55 (Tier-2 Cuttlefish driver — REAL A/B on live cvd, nezha 2026-06-23), F68, F72 (Build resource stats), F73, F77-F85, F88, F92-F93, F96 (Production Deploy), F97 (Remote Stress), F106 (§11.4.159 Recording Compliance), F108 (HelixTrack Integration), F109 (Build Resource Stats), F112 (Cuttlefish pkg/cuttlefish — REAL A/B PASS on live cvd, nezha 2026-06-23), F114 (Container stack distribution — REAL deploy to thinker GREEN: `(healthy)+/health` 200), F117 (commit_all cascade-push)
PROVEN	6	F50 (PWU-AB-1 base+boot), F51 (PWU-AB-1 slot switch), F52 (PWU-AB-3 auto-rollback), F53 (PWU-AB-2 RAUC dm-verity), F94 (AB Slot Switch Video), F95 (AB Rollback Video)
OPERATOR-BLOCKED	1	F56 (Tier-3 HW)
PARTIAL	3	F86 (stress+chaos coverage), F115 (Docker-fallback distribution — gate logic dry-run-verified, real docker

Status	Count	Items
		deploy to amber NOT run), F116 (Remote-emulator full-detachment — setsid source hardening applied + reviewed, on-target persistence NOT run-proven)
NOT_STARTED	0	

F113 (RK3588 control-plane validation) is a **PASS** for the Device-B control-plane round-trip on real RK3588 Android-15 hardware (sink-side `update_state=success`), with Device-A an honest topology SKIP — counted under PASS above. Both boards are NON-A/B (control-plane validation only; native A/B fidelity is F112/F55's Cuttlefish path — now **VERIFIED on nezha 2026-06-23** with real update_engine A/B apply + slot flip + auto-rollback, evidence `docs/qa/20260623-cuttlefish-tier2-ab/`).

Provenance

Date	Scope
2026-06-23	Rev 21 — **Cuttlefish Tier-2 REAL Android A/B VERIFIED on nezha (terminal goal met).** A real ~1 GB OTA payload was applied through `update_engine` to a live Cuttlefish cvd (build 15660610, `aosp_cf_x86_64_only_phone`, Virtual A/B + verity enforcing, 15 A/B partitions) on the operator's nezha Linux+KVM host, driven autonomously as `milosvasic` over `adb -s 127.0.0.1:6520` (no host sudo for the A/B flow): `onPayloadApplicationComplete(kSuccess)` → `UPDATE_STATUS_UPDATED_NEED_REBOOT` (115 s) → reboot slot flip `_a→_b` (VAB merge `merging→none`, `_b` marked successful) → forced-bad slot `_a` (bootctl set-slot-as-unbootable + bounded 256 KB inactive-slot boot_a write, §11.4.133) rejected → device booted known-good `_b`. OTA payload obtained with NO credentials (androidbuildinternal pre-signed GCS URL `storage.googleapis.com`, 1003473429 B, md5 `d90870a9a6eeece3868520d7fd3f098c` — size+md5 verified before apply). **F112 PARTIAL→VERIFIED; F55 (Tier-2 driver) OPERATOR-BLOCKED→VERIFIED.** Evidence + curated report `docs/qa/20260623-cuttlefish-tier2-ab/REPORT.md` (read-the-screen verified per §11.4.158). §11.4.135 regression guard `tests/regression/guard_cuttlefish_ab_proven.sh` GREEN. Validator `tests/emulator/tier2_cuttlefish_ab.sh` HONEST STATUS updated to VERIFIED + UNCONFIRMED items resolved to FACT + new running-container `--serial` mode (bash -n/sh -n clean). **Full journey (provenance):** curl-download-fail → resumable wget-c recovery → 27.6 GB single-stage image → slim 1.11 GB prebuilt-deb path

Date	Scope
	(containers `54aa9b2`) → operator privileged launch (`cf-launch.sh`) → cvd booted → this A/B PASS; cvd left running on nezha. Summary-by-Status: VERIFIED 25→27, PARTIAL 4→3, OPERATOR-BLOCKED 2→1. Honest boundary (§11.4.3/§11.4.112/§11.4.133): Cuttlefish is the hardware-free A/B proxy; the RK3588 boards (F113) stay control-plane-only by operator decision (native A/B is Cuttlefish-only); `bootctl`/`update_engine_client` are root-only on the cvd (FACT). Exports regenerated.
2026-06-18	Initial feature inventory creation (HEAD at time of writing).
2026-06-19	Feature inventory update — F90-F95 added, F88 upgraded to VERIFIED, revision 2
2026-06-19	Feature inventory update — F96-F103 added (remote deployment, stress test, security fixes, MountManagerUI, ApplyPort scaffold); Executive Summary updated; revision 3
2026-06-19	Feature inventory update — F104-F105 added (Devices List API, Hardware ID Reverse Lookup); Executive Summary updated; production E2E 288/290 noted; revision 4
2026-06-19	Recording migration + GEMINI.md lockstep — recordings moved to $HOME/Downloads, window-scoped MP4s, §11.4.159 compliance initiated; revision 5
2026-06-20	Rev 6 — PWU-AB-2 RAUC dm-verity PROVEN (GREEN 3/3 deterministic), PWU-AB-4 ApplyPort IMPLEMENTED (36 tests, 3 Go files, 2 Kotlin files, CLI binary), §11.4.159 compliance row (F106), demo re-recordings (F107), recordings count updated to 31, Summary by Status revised, all status vocabulary updated, all status vocabulary updated
2026-06-20	Rev 7 — Recording data quality fixes: count 31->30, F107 marked STALE (rotated), F94/F95 paths fixed to $HOME/Downloads/, 29 stale recordings at nonstandard path removed, Status_Summary synced | | 2026-06-20 | Rev 8 — F108 HelixTrack Integration added (DESIGN, Phase 0 committed, launcher scripts + helix-deps.yaml + space config); Phase 1-7 research agents running in parallel; Status_Summary synced | | 2026-06-20 | Rev 9 — F108 upgraded to IMPLEMENTED: Phase 1 (multi-space Core --space-root + SpaceConfig + Migrator) committed to HelixTrack. Phase 2 (docs_chain context + sync scripts) created. Phase 3 (containers compose + boot scripts) built. Phase 4 (launchers web/desktop/common) complete. Phase 5-6 (Challenge bank entries HELIXTRACK-001/002/003 + test scripts) authored. Phase 7 (Status docs) synced. Full integration pushed across all upstreams. | | 2026-06-21 | Rev 11 — F108 HelixTrack Integration row updated: 16/16 OTA items synced (not 5/5), bidirectional push/pull verified (E2E 11/11 PASS), web client Angular 19 dark-themed with brand colors (Cypress 10/10 PASS), recording evidence at /Volumes/T7/Downloads/Recordings/; exports regenerated. | | 2026-06-21 | Rev 14 — 7 new combined batch video recordings: server-recall (6 recall tests all PASS), server-auth-artifact-deploy

Date	Scope
	(auth+artifact+deploy+rollout), server-middleware-delta (middleware+delta+branches+widen), submodules-all (7 submodules, stress+chaos full PASS), server-store-stress-e2e (store+stress/chaos+coverage all PASS), frontend-security-governance (frontend+security filters+gates all PASS), server-full-suite (7 internal packages all GREEN). Total recordings: 30 → 37. 20 previously "No" Video Recorded columns updated (F04, F05, F10, F14, F20-F28, F30-F34, F40-F41, F86). Executive Summary revised. | | 2026-06-22 | Rev 16 — Durable-evidence reconciliation (§11.4.153/§11.4.158, no-bluff §11.4.6): the cleared ephemeral `$HOME/Downloadsraw corpus (§11.4.128) left ~103 video confirmations unbacked. Re-pointed the 10 autonomously-recordable server/e2e control-plane rows (F03 auth, F10/F58 recall, F12 device, F13 groups, F14 telemetry, F15 audit, F16 health, F59 rollout-halt, F60 signed-pipeline, F61 challenge-filters, F62 challenge-operational) to durable committed MP4s underdocs/qa/20260622-server-recordings-regen/mp4/ (vision-verified, REPORT.md). Re-pointed the local QEMU A/B firmware rows (F50 boot, F51/F94 slot-switch, F52/F95 rollback, F53 RAUC dm-verity) to the re-proven GREEN console evidence underdocs/qa/20260622T07*(citing the PASS runs: slot-switch 071946Z, rollback 072030Z, rauc-verity 072224Z RED_MODE=0). Honestly re-marked the hardware/operator-gated rows (F42, F43 Android agent/bridge; F55 Tier-2 Cuttlefish; F56 Tier-3 RK3588) as OPERATOR-BLOCKED — no on-host recording — citing OTA-004 + the nezha operator-attended AVD-boot evidence (qa-results/20260622T071848Z-nezha-android-ab/). No feature rows deleted (§11.4.122); only Video-confirmation cells corrected. Remaining$HOME/Downloads/…mp4citations flagged non-durable via the Executive Summary banner. | | 2026-06-22 | Rev 17 — Three new real capabilities (honest §11.4.6): **F112** Cuttlefish Tier-2 containerizedpkg/cuttlefishcvd-lifecycle wrapper (containers submodule) —go test -race30 PASS + 1 honest topology SKIP; rootful-privileged documented exception (docs/design/CUTTLEFISH_ROOTFUL_EXCEPTION.md, §11.4.161/§11.4.112). **PARTIAL / integration-pending — container path BUILT + unit-tested, NOT yet a real-A/B PASS** (nezha Linux+KVM, assets staging, privileged launch_cvd operator-gated). **F113** real RK3588 hardware control-plane validation — evidencedocs/qa/20260622-rk3588-controlplane/REPORT.md: Device B (Ethernet, serial 1acdceab90248933) PASS — device-originated register/update-check 204/telemetry 202 + sink-sideupdate_state=successon real RK3588 Android-15 hardware; Device A (Wi-Fi) honest topology SKIP

Date	Scope
	(VPN/AP-isolated). Both boards NON-A/B (single-slot, no update_engine) → control-plane validation only; ZERO device state changes. Summary-by-Status: PASS 49→50, PARTIAL 1→2. Distribution repoint (containers) now targets thinker.local (live) + amber.local (SSH-key-pending); nezha read/import-only. | | 2026-06-22 | Rev 18 — Four new infra/distribution feature rows (honest §11.4.6, infra/process features — captured evidence is the dry-run/commit log, not on-host video): **F114** container stack distribution (distribute_stack.sh— SSH + remote rootlesspodman compose, dry-run-probe-verified vs thinker.local LIVE, nezha read/import-only); **F115** §11.4.161 operator-authorized docker fallback for amber (no rootless podman — docker-only;HELIX_ALLOW_DOCKER_FALLBACK=1explicit opt-in, default-OFF rootless-or-nothing, §11.4.112 documented constraint; amber onboarded 2026-06-22 SSH key + docker); **F116** §11.4.144setsidfull SSH-session detachment for the remote AVD launch on nezha (boot PROVEN; session-detachment hardening — on-target persistence verification still operator-attended); **F117** commit_all cascade-push fix (portable mkdir lock + honest exit + per-remote fetch timeout, four-upstream fan-out). VERIFIED count 23→27. **Cuttlefish (F112) bring-up is now RUNBOOK-READY** —docs/design/CUTTLEFISH_NEZHA_RUNBOOK.mdgives the exact operator-vs-agent step split for the real nezha run; F112 stays **PARTIAL / integration-pending — NOT a real-A/B PASS** until the runbook is executed with captured slot-flip/rollback evidence. | | 2026-06-22 | Rev 20 — **F114 promoted PARTIAL → VERIFIED on REAL fully-automated GREEN evidence (§11.4.5/§11.4.69); F115/F116 stay PARTIAL (unchanged).** A non-dry-runHELIXTRACK_REMOTE_HOST=thinker.local bash scripts/distribute_stack.shran end-to-end: it BUILT the helixtrack-core image on thinker (rootless podman-compose) from the Go 1.24 Dockerfile (withcurladded for the healthcheck), brought the stack up, and a fresh container reportedpodman ps:helixtrack-core Up (healthy)+helixtrack-postgres Up (healthy), withcurl -sf http://localhost:8080/health→ 200{“status”:“ok”}(FailingStreak=0). Evidencedocs/qa/20260622-222645-distribute-thinker-FULLY-GREEN/(podman_ps.txt,health_body.txt,deploy.log,deploy_tail.txt). **Full fix chain that made the Rev-19 blockers green:** distribute_stack.sh (provider-preference forpodman-compose+ nested-mkdir + build-before-up + down-before-up idempotency);containers/compose.helixtrack.ymlhealthcheck on/

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	health(submoduledcef56d); HelixTrack Core Dockerfilegolang:1.24+ restored gutted source (3c62217/3483699) +curlin the runtime image (d0f4bfb) — this fixed the Rev-19 finding-3 go-version blocker. **Honest boundary §11.4.6/ §11.4.28:** thinker.local is the proven LIVE rootless-podman target; the helix-layer compose file in thecontainerssubmodule (dcef56d) is an operator-approved documented §11.4.28 exception; the amber docker-fallback path (F115) is NOT yet run, and F116 on-target persistence stays operator-attended — both remain PARTIAL. VERIFIED count 24 → 25; PARTIAL count 5 → 4 (F114 moved to VERIFIED). Status_Summary → rev 12. Exports regenerated. | | 2026-06-22 | Rev 19 — **Anti-bluff reconciliation (§11.4.1/§11.4.6/§11.4.153): F114/F115/F116 downgraded VERIFIED → PARTIAL; F117 stays VERIFIED (genuinely real).** A REAL (non-dry-run)distribute_stack.shdeploy to thinker.local was run and **FAILED** with 3 defects (evidencedocs/qa/ 20260622-211644-distribute-thinker/): (1) rsync nested-mkdir gap [FIXED in script]; (2) wrong compose provider — it selected the brokenpodman composeplugin instead ofpodman-compose[FIXED — now preferspodman-compose, dry-run confirms]; (3) the HelixTrack sibling Dockerfile pinsgolang:1.22butgo.modneedsgo 1.24→ helixtrack-core image build fails [NOT fixed — sibling-repo blocker;rootcause.log/go_mod_bug.log/ final_state.log]. So **NO successful end-to-end deploy has happened**, and the prior Rev-18 "VERIFIED via dry-run probe" + stale "podman composeselected" claims overstated reality. **F114 → PARTIAL** (mechanism dry-run-verified + 2 script bugs fixed, real deploy blocked on finding-3 — NOT a real-deploy PASS; corrected "podman composeselected" → "podman-composeselected after fix"). **F115 → PARTIAL** (docker-fallback gate logic + host-selection dry-run-verified, NO real docker deploy to amber executed). **F116 → PARTIAL** (setsidsource hardening applied + code-reviewed, earlier AVD boot proven, on-target persistence-after-session-end NOT run-proven / operator-attended). **F117 stays VERIFIED** — multiple real commits (e.g.17cbd47a,74af4684`) genuinely pushed to all four mirrors this session via the fix. VERIFIED count 27 → 24; PARTIAL count 2 → 5 (F114/F115/F116 added). Status_Summary → rev 11. Exports regenerated.
2026-06-21	Rev 15 — 18 new batch recordings covering all remaining PASS/VERIFIED/IMPLEMENTED features: F02 (device-emu binary), F06 (release), F11-F13 (client+device+group handlers),

Date	Scope
	F15-F16 (audit+health), F27 (config), F29 (PostgreSQL store), F30-F34 (server packages), F44-F49+F57 (emulator tiers), F54+F102 (ApplyPort), F58-F62 (e2e suites), F68+F72-F73 (containers+stats+gitignore), F74-F76 (scripts), F77+F83 (constitution+AGENTS), F78-F82 (governance docs), F84-F85+F87 (Status docs+DB), F86 (stress+chaos), F89+F110 (Docs Chain), F92-F93 (frontend), F96-F97 (deploy+stress), F98-F103 (security+deploy), F109 (resource stats), F111 (DB migration). Total recordings: 37 → 55. Executive Summary revised, Revision bumped.